

AI and Applied Math circa September 2026

Excitement, ethics, and individual exploration

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Why this talk

- 8 September: OpenAI says its AI settled a 90-year-old, \$1M question about fluids.
- The night before: two mathematicians, working with Claude and Codex, go public with a year of closely related work.
- Two weeks ago: I rebuilt the simulation methods from my 2016 dissertation with Claude Fable 5.1, in a day.

Thesis

AI agents now bring a staggering amount of programming and logical power. Paired with a human's experience and intuition in a field, they make progress fast, at every scale: 10,000 agents on a Millennium problem, or one assistant in my corner of PDEs.

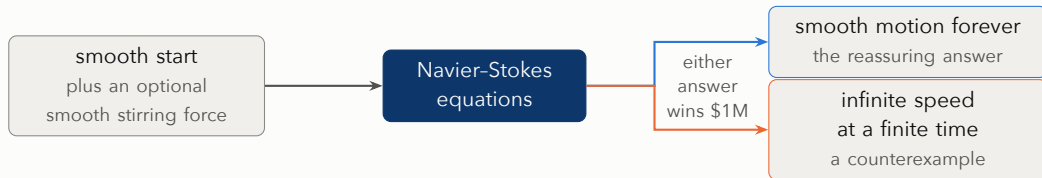
Part 1 (15 min). The Navier-Stokes claim: what was claimed, how it was produced, who else was there, what is still open.

Part 2 (15 min). Wave simulations from my dissertation, re-derived with an assistant. Two short videos.

Part 1: the Navier-Stokes claim

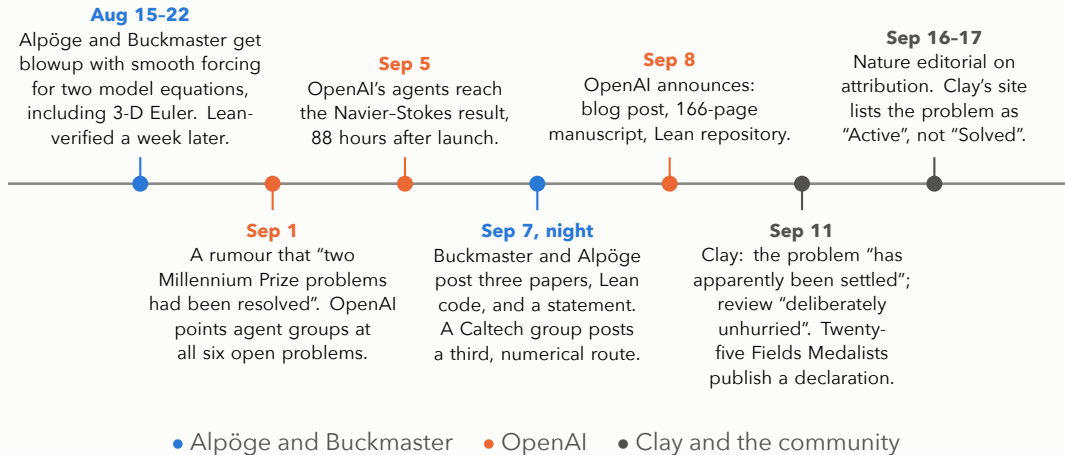
Fluids, Newton, and a question from 1934

- The **Navier-Stokes equations** are Newton's $F = ma$ for a fluid: water, air, blood. They sit inside aircraft design, weather forecasting, and ocean models.
- Leray's question (1934): from a smooth start, do the equations always predict smooth motion, or can they predict **infinite speed in finite time** ("blowup")? Real fluids cannot; blowup would mean the model leaves its own range of validity.
- Clay Mathematics Institute, 2000: \$1M for an answer either way.



Source: C. Fefferman, *Existence and smoothness of the Navier-Stokes equation*, Clay Mathematics Institute, p. 2.

What happened, in two weeks

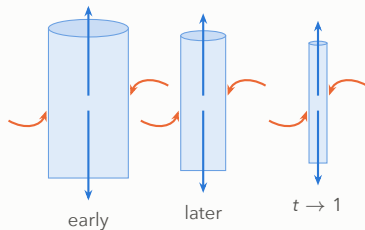


Sources: OpenAI's post; Buckmaster's statement; claymath.org; mathandai.org; nature.com. Every date is traced in docs/navier-stokes-notes.md.

The claim, and the picture behind it

Abstract, verbatim

"For every positive viscosity, we construct a solution of the three-dimensional incompressible Navier-Stokes equations that starts from rest and develops unbounded velocity in finite time while maintaining uniformly bounded kinetic energy."



After Fig. 1 of the manuscript. Orange:
inward spiral; blue: outflow along the axis.

- Starts **at rest**; a smooth stirring force drives it. Energy stays bounded, yet the top speed reaches infinity at $t = 1$.
- A spinning column gets thinner, longer, faster ("like spaghetti"). Pulling mass inward speeds the spin; viscosity loses.
- The force holding the bare vortex together would itself blow up; fine **ripples** are added whose own motion cancels the bad part.
- **What it is not.** Nobody has shown that turbulence on its own blows up. The unforced question is untouched.

How it was produced, in OpenAI's own account

~10,000

concurrent agents

88 h

from launch to result

130 B

output tokens

17 h

more, to formalize in
Lean

millions

of dollars, per Mark
Chen to *Science*

- An internal model “significantly more capable than GPT-6 Astra”, in training since 28 August.
- Groups of **coordinating agents** with tools (a cached copy of the internet, code execution), each group given one of the four official variants of the problem.
- Warm-up: the unforced Euler question fell first, to “nearly 100 agents” in about 50 hours. Everything was then pointed at Navier-Stokes, seeded with that solution.
- OpenAI will not claim the prize.

Source: OpenAI's post (8 Sep, updated 10 Sep); Science, 8 Sep. Nature and Science put the warm-up at 1,000 agents; the post says “nearly 100”. Check press numbers against the primary source.

The parallel story, and the dispute: who gets credit?

- Tristan Buckmaster (NYU) and Levent Alpöge (Anthropic), with Claude and Codex, for most of a year, on Buckmaster's own research funds. Both they and OpenAI build on Córdoba and Martínez-Zoroa.
- Results 15 August, Lean-verified 22 August, released 7 September, earlier than planned. Buckmaster, on their own papers: "The Euler writeup, in particular, can only be described as AI slop."

Buckmaster's statement (7 Sep)

- Emailed OpenAI on 3 Sep; two calls on 6 Sep. Told the proof came with "very little human input", which "turned out not to be true".
- "I asked whether the model had been trained on, or had access to, our sessions in Codex ... I did not get an answer."
- Declined two proposals, one a paper by Buckmaster alone crediting an OpenAI model, with Alpöge removed.

OpenAI's post (8 Sep, updated 10 Sep)

- The effort "began on September 1st after hearing a rumor"; they "did not see any of their work through any means until they released it publicly".
- 8 Sep: "we cannot rule out that de-identified data ... helped improve our models"; 10 Sep, after an investigation: "could not have influenced the system in any way, including through training".
- "We recognize the priority of their work on forced Euler."

Neither account is independently verified: claims and responses, not findings. Sources: cims.nyu.edu/~tristanb; openai.com.

What “machine-checked” does and does not mean

Lean checks

- Lean itself: a programming language for mathematics. A proof is a program; the compiler accepts it or rejects it.
- That every step follows from three standard axioms. No gaps, no “clearly”.
- Reproducibly: clone the repository, run the independent checker.

Humans still check

- That the formal theorem *is* the Clay statement. OpenAI targeted an independent transcription of it (Google DeepMind’s). Still a judgment.
- Whether the construction teaches anything. Clay: publication, a two-year wait, then a committee.

Quanta, 8 Sep

The “crucial bit of verification that must still be done by humans is to guarantee that the statement being shown to be true in Lean is logically equivalent to what mathematicians set out to prove.”

Reactions, and questions to argue about

Reactions

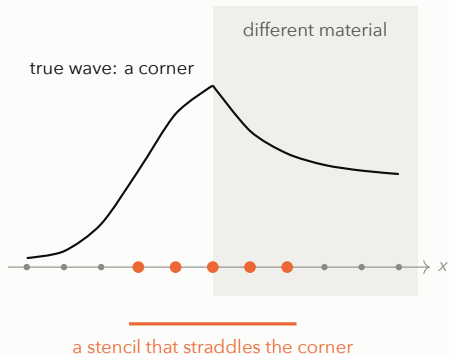
- **Clay Mathematics Institute** (11 Sep): “apparently been settled”; evaluation “deliberately unhurried”. Site status: *Active*.
- **25 Fields Medalists** (11 Sep): “solving problems is only a tool and proxy for achieving the primary goal of conceptual understanding and insight”.
- **Terence Tao**: open problems “are being mined as a non-renewable resource”.
- **Tristan Buckmaster**: “This is a Deep Blue-Kasparov moment.”
- **Sébastien Bubeck** (OpenAI): “the spectacular culmination of the arc we have seen over the last 12 months”.

Questions

1. **Definition of done.** The statement allows a smooth force, and this result uses one. Is the Millennium problem solved?
2. **Trust.** Would you put unpublished work into a frontier lab’s tools? What would you need to see first?
3. **Compute.** No university can run 10,000 agents for four days. What does that do to a field whose scarce resource was always ideas?

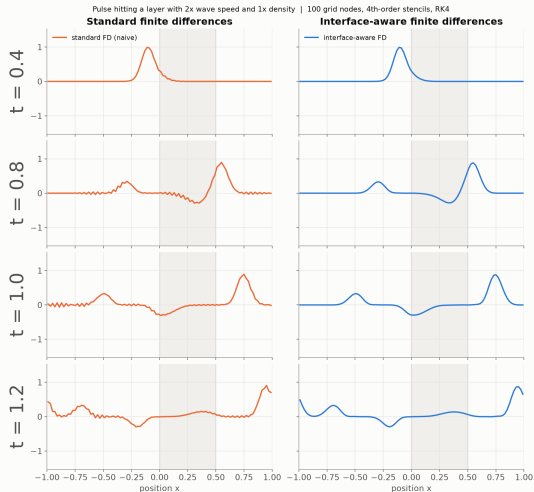
Part 2: my end of the scale

My corner of this world: waves through layered materials



- 2016 PhD, applied mathematics, CU Boulder: seismic imaging (sound through rock), ultrasound (sound through tissue).
- A simulation stores the wave at grid points and estimates slopes from a few neighbours: a **stencil**. One that straddles the corner assumes a smooth curve and gets the slope wrong.
- The fix: rebuild the few stencils near the boundary from pieces that are allowed a corner there. Nothing else changes.
- The experiment: nine-year-old MATLAB, no licence. With Claude Fable 5.1: re-derive from the papers, port to Python, verify against exact solutions. One day.

1-D: the ringing, and the fix

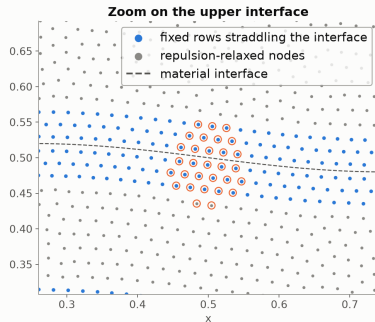
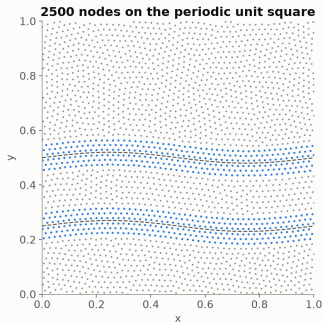


- A pulse hits a layer (shaded) where waves travel twice as fast. Part reflects, part transmits.
- Same 100 grid points, same time step, same code. Only the stencils touching the layer edges differ.
- **Standard**: wiggles behind every reflection; 10% error by $t = 1$.
Interface-aware: 3% error at $t = 1$, all of it from the coarse grid.

Play clip 1 (10 s)

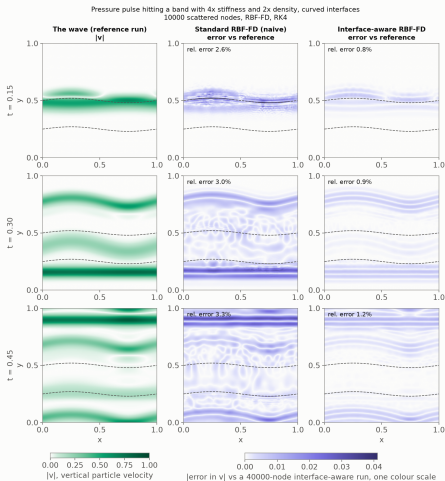
wave1d_naive_vs_aware_coarse.mp4

2-D: scattered points and curved boundaries



- No grid. Points are spread by a repulsion process, with fixed rows straddling each boundary (blue). That layout was the key to stability: years to find, still unproven. (2,500 shown; the runs use 10,000.)
- Each point's stencil is its 30 nearest neighbours, weighted with radial basis functions (RBF-FD). Five fields: two velocities, three stresses.

2-D: same points, only the stencils near the boundaries differ



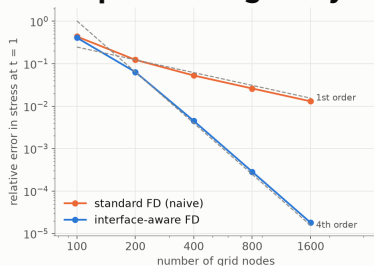
- A pressure pulse travels down through a curved band of stiffer, denser material. 10,000 points.
- Left: **the wave**, from a 40,000-point reference run. Right pair: each method's **error** against it, one colour scale.
- At $t = 0.3$: **standard** 3.0% error, **interface-aware** 0.9%. The standard error is born at the boundary and travels with the wave.

Play clip 3 (8 s)

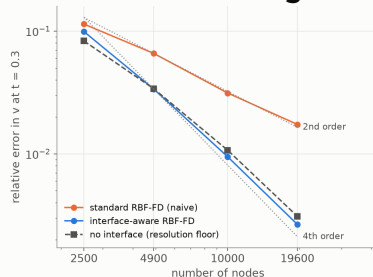
wave2d_naive_vs_aware_curved.mp4

How fast the error shrinks

1-D: a pulse through a layer



2-D: elastic waves through a band



- **Standard**: double the points, halve the error, in 1-D and 2-D alike. The corner caps the accuracy.
- **Interface-aware**: double the points, divide the error by 16 (1-D) or by 4 (2-D).
- Same story for a thinner layer.

Python port of dissertation Figs. 2-8 and 3-5, error against exact solutions; thin-layer and curved cases in the repository.

What the collaboration looked like

1

day (17 September)

9

pull requests

106

tests, all green

4

real bugs caught
before the demo

2

methods, 1-D and
2-D

Fable 5.1 assistant

- Every routine tested against an exact solution or a convergence rate, not “it runs”.
- Independent reference solutions built before trusting a solver.
- Review and cross-checks caught a sign error, two latent bugs in the exact solver that only the 2-D problem exposed, and a driver crash.

Still human

- Where to put the points (the straddling rows). Plain method first, clever method second.
- Which basis functions, and the narrow window for the stabilising parameter.
- Steering, judging what matters, designing the verification. The typing, the bookkeeping, and the port were not.

Two scales of AI collaboration in research math

	OpenAI, Navier-Stokes	Alpöge and Buckmaster	Me, with Claude Fable 5.1
Who steered	a lab team, via prompts	two mathematicians	one engineer
Scale	~10,000 agents, 88 h	about a year, several models	one agent, one day
Verification	Lean; Clay review pending	Lean; papers being rewritten	tests against exact solutions
Who understands it	contested	"AI slop", being digested	fully; it is my method
Cost	millions of dollars	own research funds	\$200/mo 20x Max subscription

- The capability is impressive at all scales in this table.
- The difference is who steered, who can check it, and who is left understanding the result.

For the curious

Gentle starts

- 3Blue1Brown, [But what is a partial differential equation?](#) (YouTube, 17 min)
- Khan Academy: [Differential equations](#)
- Wikipedia: [Navier-Stokes existence and smoothness](#); [Finite difference method](#); [Lean](#) (proof assistant)

Primary sources for Part 1

- OpenAI: [the announcement and manuscript](#); [the Lean repository](#)
- Clay Mathematics Institute: [the problem page and official statement](#)
- [Buckmaster's statement](#) and papers (cims.nyu.edu/~tristanb)
- [Tao's blog post, 7 Sep 2026](#); [the Fields Medalists' declaration](#)

Part 2

- Code, tests, figures, the equations, and these notes:
github.com/bradleyymartin/20260930-zd-ai-pdes-demo (MIT)